Empowerment through quality technical education
AJEENKYA DY PATIL SCHOOL OF ENGINEERING
"Imparting quality education in the field of Artificial Intelligence and Data Science"
Department of Artificial Intelligence & Data Science Engineering
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IQAC/12a

LAB MANUAL

Software Laboratory III

(317534)

TE (AI&DS) 2025-26

Course Coordinator

Prof. Neelam Jain

DEPARTMENT OF

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Department of Artificial Intelligence & Data Science

Vision:

Imparting quality education in the field of Artificial Intelligence and Data Science

Mission:

- To include the culture of R and D to meet the future challenges in AI and DS.
- To develop technical skills among students for building intelligent systems to solve problems.
- To develop entrepreneurship skills in various areas among the students.
- To include moral, social and ethical values to make students best citizens of country.

Program Educational Outcomes:

1. To prepare globally competent graduates having strong fundamentals, domain knowledge, updated with modern technology to provide the effective solutions for engineering problems.

2. To prepare the graduates to work as a committed professional with strong professional ethics and values, sense of responsibilities, understanding of legal, safety, health, societal, cultural and environmental issues.

3. To prepare committed and motivated graduates with research attitude, lifelong learning, investigative approach, and multidisciplinary thinking.

4. To prepare the graduates with strong managerial and communication skills to work effectively as individuals as well as in teams.

Program Specific Outcomes:

1. Professional Skills- The ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, networking, artificial intelligence and data science for efficient design of computer-based systems of varying complexities.

2. Problem-Solving Skills- The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for business success.

3. Successful Career and Entrepreneurship- The ability to employ modern computer languages, environments and platforms in creating innovative career paths to be an entrepreneur and to have a zest for higher studies.

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1. Guidelines to manual usage

This manual assumes that the facilitators are aware of collaborative learning methodologies.

This manual will provide a tool to facilitate the session on Digital Communication modules in collaborative learning environment.

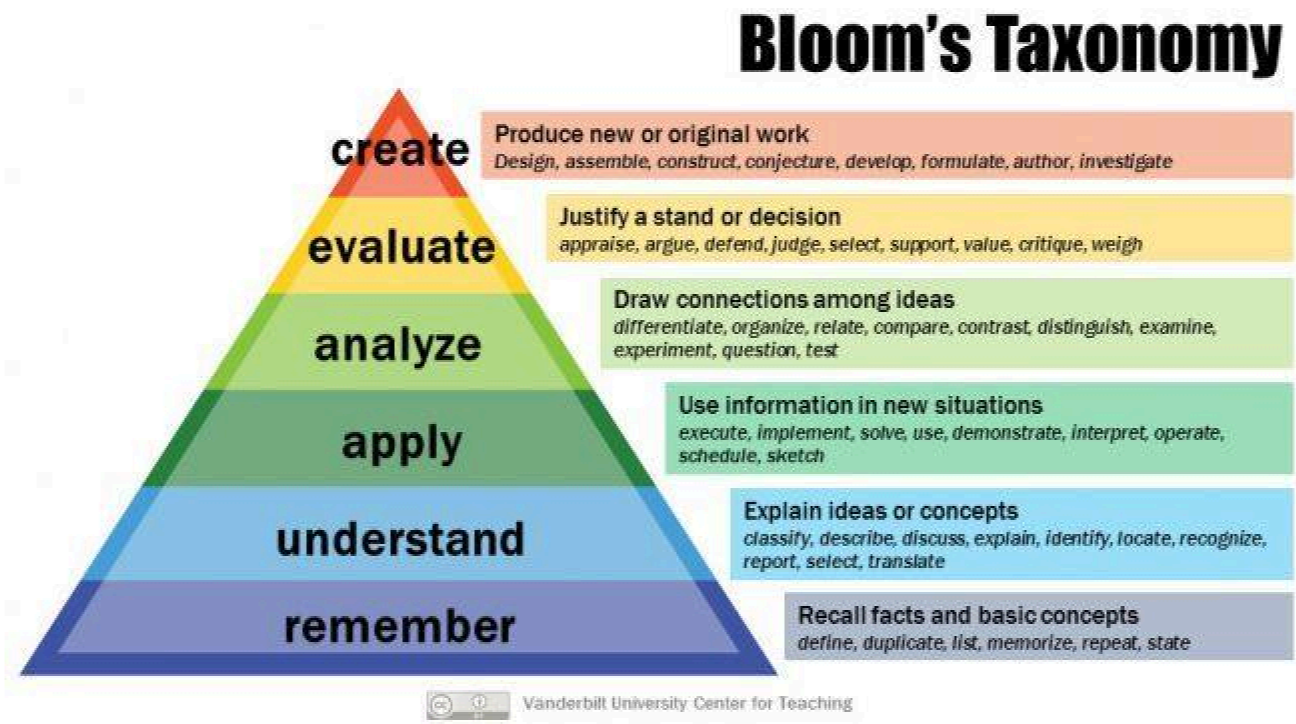
The facilitator is expected to refer this manual before the session.

Icon of Graduate Attributes

K Applying Knowledge	**A** Problem Analysis	**D** Design & Development	**I** Investigation of problems
M Modern Tool Usage	**E** Engineer & Society	**E** Environment Sustainability	**T** Ethics
T Individual & Team work	**O** Communication	**M** Project Management & Finance	**I** Life-Long Learning

Blooms Taxonomy

Bloom's Taxonomy is a classification of the different objectives and skills that educators set for their students (learning outcomes).



Program Outcomes:

PO1: Engineering knowledge: An ability to apply knowledge of mathematics, including discrete mathematics, statistics, science, computer science and engineering fundamentals to model the software application.

PO2: Problem analysis: An ability to design and conduct an experiment as well as interpret data, analyze complex algorithms, to produce meaningful conclusions and recommendations.

PO3: Design/development of solutions: An ability to design and development of software system, component, or process to meet desired needs, within realistic constraints such as economic, environmental, social, political, health & safety, manufacturability, and sustainability.

PO4: Conduct investigations of complex problems: An ability to use research based knowledge including analysis, design and development of algorithms for the solution of complex problems interpretation of data and synthesis of information to provide valid conclusion.

PO5: Modern tool usage: An ability to adapt current technologies and use modern IT tools, to design, formulate, implement and evaluate computer based system, process, by considering the computing needs, limits and constraints.

PO6: The engineer and society: An ability of reasoning about the contextual knowledge of the societal, health, safety, legal and cultural issues, consequent responsibilities relevant to IT practices.

PO7: Environment and sustainability: An ability to understand the impact of engineering solutions in a societal context and demonstrate knowledge of and the need for sustainable development.

PO8: Ethics: An ability to understand and commit to professional ethics and responsibilities and norms of IT practice.

PO9: Individual and team work: An ability to apply managerial skills by working effectively as an individual, as a member of a team, or as a leader of a team in multidisciplinary projects.

PO10: Communication: An ability to communicate effectively technical information in speech, presentation, and in written form

PO11: Project management and finance: An ability to apply the knowledge of Information Technology and management principles and techniques to estimate time and resources needed to complete engineering project.

PO12: Life-long learning: An ability to recognize the need for, and have the ability to engage in independent and life-long learning.

Course Name: Software Laboratory III

Course Code: (317534)

Course Outcomes

CO1: Apply principles of Data Science for the analysis of real time problems
CO2: Implement data representation using statistical methods
CO3: Implement and evaluate data analytics algorithms
CO4: Perform text preprocessing
CO5: Implement data visualization techniques
CO6: Use cutting edge tools and technologies to analyze Data

CO to PO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	2	2	`	`	`	`	`	`
CO2	2	2	2	2	3	-	`	`	`	`	`	`
CO3	2	2	2		2	-	`	`	`	`	`	`
CO4	2	2	2	2	2	2	`	`	`	`	`	`
CO5	2	2	2	2	2	2	`	`	`	`	`	`
CO6	2	2	2	2	2	2	`	`	`	`	`	`

CO to PSO Mapping:

	PSO1	PSO2	PSO3
CO1			
CO2			
CO3			
CO4			
CO5			
CO6			

2. Laboratory Objective

● To understand principles of Data Science for the analysis of real time problems
● To develop in depth understanding and implementation of the key technologies in Data Science and Data Analytics
● To analyze and demonstrate knowledge of statistical data analysis techniques for decision-making
● To gain practical, hands-on experience with statistics programming languages and Data tools

3. Laboratory Equipment/Software:

The laboratory is equipped with the following hardware and software resources to support the practical implementation of data science and machine learning concepts:

Hardware

- Desktop computers / Laptops with a minimum **Intel i5 / Ryzen 5 processor**
- Minimum **8 GB RAM**
- Internet connectivity
- Projector / Smart Board (for demonstrations)

Software

- **Operating System:** Windows / Linux
- **Programming Language:** Python (3.x)
- **IDE / Tools:**
  - Jupyter Notebook
  - Anaconda Distribution
  - Google Colab (cloud-based)
- **Libraries & Frameworks:**
  - NumPy
  - Pandas
  - Matplotlib
  - Seaborn
  - Scikit-learn

4.Laboratory Experiment List

Sr. No	Title
	Prerequisite practical assignments or installation (if any)
1	Basic understanding of relational database concepts, SQL queries, and normalization
2	Installation and familiarity with any one DBMS such as MySQL / Oracle / PostgreSQL
	List of Assignments
1	**Data Wrangling, I** Perform the following operations using Python on any open-source dataset (e.g., data.csv) 1. Import all the required Python Libraries. 2. Locate open source data from the web (e.g., https://www.kaggle.com). Provide a clear description of the data and its source (i.e., URL of the web site). 3. Load the Dataset into pandas dataframe. 4. Data Preprocessing: check for missing values in the data using pandas isnull (), describe() function to get some initial statistics. Provide variable descriptions. Types of variables etc. Check the dimensions of the data frame. 5. Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions. 6. Turn categorical variables into quantitative variables in Python. In addition to the codes and outputs, explain every operation that you do in the above steps and explain

	everything that you do to import/read/scrape the data set
2	**Data Wrangling II** Create an "Academic performance" dataset of students and perform the following operations using Python. 1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them. 2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them. 3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution. Reason and document your approach properly
3	**Descriptive Statistics** Measures of Central Tendency and variability Perform the following operations on any open source dataset (e.g., data.csv) 1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a Curriculum for Third Year of Artificial Intelligence and Data Science (2019 Course), Savitribai Phule Pune University http://collegecirculars.unipune.ac.in/sites/documents/Syllabus2022/Forms/AllItems.asp x #84/105 dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variable. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable. 2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset.
4	**Data Analytics I** Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset (https://www.kaggle.com/c/boston-housing). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset. The objective is to predict the value of prices of the house using the given features.
5	**Data Analytics II** 1. Implement logistic regression using Python/R to perform classification on Social_Network_Ads.csv dataset. 2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.
6	**Data Analytics III**

	1. Implement Simple Naïve Bayes classification algorithm using Python/R on iris.csv dataset. 2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.
7	**Text Analytics 1**. Extract Sample document and apply following document preprocessing methods: Tokenization, POS Tagging, stop words removal, Stemming and Lemmatization. 2. Create representation of documents by calculating Term Frequency and Inverse DocumentFrequency.
8	**Data Visualization I** 1. Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about the passengers who boarded the unfortunate Titanic ship. Use the Seaborn library to see if we can find any patterns in the data. 2. Write a code to check how the price of the ticket (column name: 'fare') for each passenger is distributed by plotting a histogram.
9	**Data Visualization II** 1. Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names: 'sex' and 'age') 2. Write observations on the inference from the above statistics.
10	**Data Visualization III** Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as: 1. List down the features and their types (e.g., numeric, nominal) available in the dataset. 2. Create a histogram for each feature in the dataset to illustrate the feature distributions. 3. Create a boxplot for each feature in the dataset. 4. Compare distributions and identify outliers.
11	Create databases and tables, insert small amounts of data, and run simple queries using Impala
12	Write a simple program in SCALA using Apache Spark framework
	Content Beyond Syllabus

4.1 Experiment No. 1 — Data Wrangling I

Aim:

To perform data wrangling operations using Python on any open-source dataset (e.g., Kaggle dataset such as Titanic.csv).

Objective:

To study:

- Importing and reading open-source datasets using Python.
- Identifying and handling missing values and data inconsistencies.
- Formatting, normalizing, and converting categorical data into numerical data.

Theory:

Data wrangling (or preprocessing) is the process of converting raw data into clean, structured, and analyzable form. Real-world datasets often contain missing values, incorrect datatypes, and categorical variables that must be transformed for analysis.
Python libraries such as **NumPy**, **Pandas**, and **Scikit-learn** provide tools for:

- Handling missing values
- Encoding categorical variables
- Normalizing data
- Type conversion

Applications:

- Preparing datasets for machine learning models
- Cleaning enterprise datasets before visualization
- Converting categorical data to usable numeric form
- Ensuring data quality for analytics pipelines

Input:

Any open-source dataset downloaded from the web. Example: Titanic dataset with passenger age, fare, class, gender, etc.

Output:

A cleaned dataset where:

- Missing values are identified and handled
- Datatypes are corrected
- Categorical data is encoded into numerical form

Conclusion:

Data wrangling was successfully performed using Python. The dataset was explored, cleaned, and transformed by handling missing values and converting categorical features for further analysis.

Outcome (ELO):

- **ELO1:** Import and describe datasets using Pandas/NumPy
- **ELO2:** Identify and handle missing values and datatypes
- **ELO3:** Encode categorical variables into numerical form

Viva Questions:

1. What is data wrangling?
2. Why is datatype conversion important in preprocessing?
3. What is One-Hot Encoding?
4. Why do we normalize data?
5. Which Python libraries are used for data preprocessing?

4.2 Experiment No. 2 — Data Wrangling II (Academic Performance Dataset)

Aim:

To create an Academic Performance dataset and perform preprocessing operations including handling missing values, outlier detection, and transformations.

Objective:

To study:

- Detecting missing values and inconsistencies
- Handling outliers
- Performing transformations to reduce skewness or improve interpretability

Theory:

Real datasets often contain irregularities such as:

- Missing values
- Typographical errors
- Extreme outliers
- Skewed distributions

Common techniques include:

- **Imputation:** mean, median, mode replacement
- **Outlier treatment:** capping, removal, Winsorization
- **Transformations:** log, sqrt, z-score standardization, min-max scaling

Applications:

- Improving model accuracy by eliminating distortions
- Making variable distributions normal
- Creating high-quality datasets for statistical analysis

Input:

A manually created "Academic Performance" dataset with fields like:

- Student Name
- Marks
- Attendance
- Grade
- Study Hours

Output:

A cleaned dataset with:

- Missing values handled
- Outliers identified and treated
- At least one transformed variable (e.g., log-transformed study hours)

Conclusion:

All preprocessing steps were successfully carried out, including missing value treatment, outlier detection, and data transformation for better analysis.

Outcome (ELO):

- **ELO1:** Handle missing values
- **ELO2:** Detect and treat outliers
- **ELO3:** Apply data transformation and justify the technique used

Viva Questions:

1. What are outliers?
2. What is Winsorization?
3. Why are transformations applied?
4. What is z-score normalization?
5. How do you identify skewness?

4.3 Experiment No. 3 — Descriptive Statistics

Aim:

To compute descriptive statistics and perform grouped analysis on a dataset using Python.

Objective:

To study:

- Measures of central tendency
- Measures of dispersion
- Group-wise statistical analysis
- Descriptive statistics of datasets like iris.csv

Theory:

Descriptive statistics summarize the essential characteristics of a dataset:

- **Central tendency:** mean, median, mode
- **Dispersion:** variance, standard deviation, range
- **Percentiles and quartiles:** measure distribution spread
  Python modules like Pandas offer functions such as:
- describe()
- group by()
- quantile ()

Applications:

- Understanding dataset behavior
- Summarizing statistical properties
- Feature selection and understanding variability
- Detecting anomalies

Input:

Any numeric dataset (e.g., iris.csv, income dataset, age groups).

Output:

- Summary statistics table
- Grouped statistics (e.g., income grouped by age bracket)
- Descriptive statistics for Iris species

Conclusion:

Descriptive statistics and grouped analysis were successfully performed. Percentiles, standard deviation, and mean values helped understand dataset patterns.

Outcome (ELO):

- **ELO1:** Compute descriptive statistics in Python
- **ELO2:** Generate grouped summaries
- **ELO3:** Interpret results and observe insights

Viva Questions:

1. What is the difference between variance and standard deviation?
2. What are quartiles?
3. What is the purpose of using grouped statistics?
4. Define percentile.
5. What does describe() return?

4.4 Experiment No. 4 — Data Analytics I (Linear Regression)

Aim:

To create a Linear Regression model using Python to predict house prices using the Boston Housing Dataset.

Objective:

To study:

1. Performing supervised learning using Linear Regression.
2. Understanding relationships between independent and dependent variables.
3. Evaluating regression model performance using metrics.

Theory:

Linear regression is a supervised machine learning technique used to model the linear relationship between:

- **Independent variables (features)**
- **Dependent variable (target)**

The general equation is:

$$Y = a + bX$$

Where:

- Y = predicted value
- X = input variable
- a = intercept
- b = coefficient

The **Boston Housing dataset** contains 506 entries and 14 attributes including crime rate, number of rooms, property tax, and more. The goal is to predict **MEDV** (Median Home Value).

Python's scikit-learn library provides tools for:

- Model creation

- Training/testing split
- Performance evaluation

Applications:

- Real estate price prediction
- Financial forecasting
- Risk assessment
- Cost estimation

Input:

Boston Housing dataset downloaded from Kaggle or accessed from sklearn datasets.

Output:

- Trained Linear Regression model
- Coefficients of features
- Prediction of house prices
- Model performance metrics such as R^2 score, MAE, and MSE

Conclusion:

A linear regression model was successfully developed to predict housing prices. Model performance metrics helped understand how accurately the features explain the variations in house values.

Outcome (ELO):

- **ELO1:** Understand supervised learning concepts
- **ELO2:** Build and train a Linear Regression model
- **ELO3:** Evaluate regression performance using metrics

Viva Questions:

1. What is linear regression?
2. What is the significance of R^2 score?
3. What is the difference between MAE and MSE?

4. What does a regression coefficient represent?
5. Why do we split data into train and test sets?

4.5 Experiment No. 5 — Data Analytics II (Logistic Regression & Confusion Matrix)

Aim:

To implement Logistic Regression for classification using Python on Social_Network_Ads.csv and compute the confusion matrix (TP, FP, TN, FN, Accuracy, Precision, Recall).

Objective:

To study:

1. Binary classification using Logistic Regression.
2. Evaluating classifier performance using confusion matrix.
3. Understanding performance metrics like precision & recall.

Theory:

Logistic Regression is a classification algorithm that predicts categorical outcomes (e.g., Yes/No, 0/1).
It uses the **sigmoid function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The **confusion matrix** explains classifier performance:

Term	Meaning
TP	True Positive
FP	False Positive
TN	True Negative
FN	False Negative

Performance measures:

- **Accuracy** = (TP + TN) / Total
- **Precision** = TP / (TP + FP)

- **Recall** = TP / (TP + FN)
- **Error Rate** = 1 – Accuracy

Applications:

- Customer churn prediction
- Medical diagnosis
- Email spam detection
- Fraud detection

Input:

Social_Network_Ads.csv with fields such as Age, Estimated Salary, Purchased.

Output:

- Logistic Regression classifier model
- Confusion matrix values (TP, FP, TN, FN)
- Performance metrics (Accuracy, Precision, Recall)

Conclusion:

Logistic Regression was successfully applied to classify user purchasing behavior. Confusion matrix and derived metrics quantified model accuracy and classification performance.

Outcome (ELO):

- **ELO1:** Implement logistic regression for binary classification
- **ELO2:** Create and interpret confusion matrix
- **ELO3:** Compute and analyze model evaluation metrics

Viva Questions:

1. Why is Logistic Regression used for classification?
2. What is a confusion matrix?
3. Define precision and recall.
4. What is the sigmoid function used for?

5. What is the difference between logistic and linear regression?

4.6 Experiment No. 6 — Data Analytics III (Naïve Bayes Classifier)

Aim:

To implement the Simple Naïve Bayes classification algorithm using Python on iris.csv and compute confusion matrix metrics.

Objective:

To study:

1. Probabilistic classification using Naïve Bayes.
2. Applying Bayes theorem on real datasets.
3. Computing TP, FP, TN, FN and evaluation metrics.

Theory:

Naïve Bayes classifier applies **Bayes' Theorem**:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

It assumes **independence** between features — hence the name *Naïve*.
Types include:

- Gaussian NB
- Multinomial NB
- Bernoulli NB

Iris dataset contains 150 samples and 3 species:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Features: sepal length, petal width, etc.

Applications:

- Document classification
- Sentiment analysis
- Email spam filtering
- Medical diagnosis prediction

Input:

Iris dataset (iris.csv)

Output:

- Naïve Bayes trained model
- Class predictions
- Confusion matrix & metrics: Accuracy, Precision, Recall, Error Rate

Conclusion:

Naïve Bayes classification was successfully implemented on the iris dataset. The model performed well for multi-class classification, and performance metrics validated its accuracy.

Outcome (ELO):

- **ELO1:** Understand probabilistic classification
- **ELO2:** Implement Naïve Bayes classifier
- **ELO3:** Interpret confusion matrix and model performance

Viva Questions:

1. What is Bayes' Theorem?
2. Why is the algorithm called "Naïve"?
3. What is the assumption behind Naïve Bayes?
4. What are types of Naïve Bayes classifiers?
5. Why is Naïve Bayes good for text classification?

4.7 Experiment No. 7 — Text Analytics (Preprocessing + TF-IDF)

Aim:

To perform text preprocessing operations such as tokenization, POS tagging, stop-word removal, stemming, lemmatization, and to compute Term Frequency and Inverse Document Frequency (TF-IDF) for the given documents.

Objective:

To study:

1. Text preprocessing techniques.
2. Tokenization and Part-of-Speech (POS) tagging.
3. Stop-word removal, stemming, and lemmatization.
4. Document representation using TF and TF-IDF.

Theory:

Text analytics converts raw text into meaningful structured information.
Common preprocessing steps include:

- **Tokenization:** Splitting text into words/tokens.
- **POS Tagging:** Identifying the grammatical role of each word.
- **Stop-word Removal:** Removing frequently occurring words (e.g., the, is, and).
- **Stemming:** Reducing words to their base form (e.g., "playing" → "play").
- **Lemmatization:** Converting words to their root using vocabulary (e.g., "better" → "good").

TF-IDF is used to identify important words in a document.

- **Term Frequency (TF):**

$$TF = \frac{\text{No. of occurrences of a word}}{\text{Total words}}$$

- **Inverse Document Frequency (IDF):**

$$IDF = \log\left(\frac{N}{df}\right)$$

- **TF-IDF:**

$$TF\text{-}IDF = TF \times IDF$$

Applications:

- Search engines
- Spam detection
- Sentiment analysis
- Chatbots
- Document classification

Input:

A sample text document or paragraph.

Output:

- Tokenized text
- POS-tagged output
- Cleaned text after removing stop-words
- Stemmed and lemmatized output
- TF and TF-IDF values for tokens

Conclusion:

Text preprocessing was successfully performed. The document was tokenized, cleaned, and transformed, and TF-IDF scores were calculated to identify important terms.

Outcome (ELO):

- **ELO1:** Apply text preprocessing using NLTK.
- **ELO2:** Perform tokenization, POS tagging, stemming, and lemmatization.

- **ELO3:** Compute TF and TF-IDF values.

Viva Questions:

1. What is the difference between stemming and lemmatization?
2. Why are stop-words removed?
3. What does TF-IDF represent?
4. What is POS tagging?
5. Why do we preprocess text?

4.8 Experiment No. 8 — Data Visualization I (Titanic Dataset, Seaborn)

Aim:

To visualize the Titanic dataset using Seaborn and identify patterns. To analyze the distribution of ticket fare using a histogram.

Objective:

To study:

1. Data visualization using Seaborn.
2. Histogram plotting and distribution analysis.
3. Pattern detection in the Titanic dataset.

Theory:

Data visualization helps in understanding hidden patterns and trends.
The **Seaborn** library provides advanced plots such as:

- Histograms
- Count plots
- Bar plots
- Box plots
- Pair plots

The Titanic dataset includes:

- Passenger class
- Gender
- Age
- Fare
- Survival status

Histograms help visualize the **distribution** of numeric variables like fare.

Applications:

- Detecting skewness in data

- Understanding demographic behavior
- Identifying factors influencing survival
- Exploratory Data Analysis (EDA)

Input:

Inbuilt Titanic dataset from Seaborn or downloaded Titanic.csv.

Output:

- Visualizations (e.g., count plots revealing survival patterns)
- Histogram showing fare distribution

Conclusion:

The Titanic dataset was successfully analyzed using Seaborn. The histogram revealed how ticket fares were distributed, and visualizations helped identify meaningful patterns such as fare variation across passengers.

Outcome (ELO):

- **ELO1:** Perform basic visualizations with Seaborn.
- **ELO2:** Analyse distributions using histograms.
- **ELO3:** Interpret patterns using visual insights.

Viva Questions:

1. What is the purpose of a histogram?
2. Which library is better for statistical visualization—Matplotlib or Seaborn, and why?
3. What patterns can be observed in Titanic survival data?
4. What is the difference between countplot and barplot?
5. What is skewness in a distribution?

4.9 Experiment No. 9 — Data Visualization II (Box Plot on Titanic Dataset)

Aim:

To plot a box plot for distribution of age with respect to gender and survival status in the Titanic dataset.

Objective:

To study:

1. Box plots for visualizing distribution and outliers.
2. Relationship between categorical and numerical variables.
3. Comparing age distribution across gender and survival groups.

Theory:

A **box plot** represents statistical summaries using five-number summary:

- Minimum
- 1st Quartile (Q1)
- Median
- 3rd Quartile (Q3)
- Maximum

It also displays **outliers** outside 1.5×IQR.

In the Titanic dataset, analyzing **age** with respect to **gender** and **survival** helps understand:

- Age groups more likely to survive
- Gender differences in survival
- Presence of outliers in age

Applications:

- Comparing age distributions in groups
- Detecting outliers
- Statistical profiling

- Gender-based survival pattern analysis

Input:

Titanic dataset (inbuilt or CSV version)

Output:

- Box plot comparing age distribution for:
  - o Male vs Female
  - o Survived vs Not Survived

Conclusion:

A box plot was successfully generated to visualize age distribution across gender and survival status. Observations revealed differences in median ages and presence of outliers in certain groups.

Outcome (ELO):

- **ELO1:** Create and interpret box plots.
- **ELO2:** Compare distributions across categorical variables.
- **ELO3:** Identify outliers through visualization.

Viva Questions:

1. What does a box plot show?
2. What is IQR?
3. Why do outliers appear?
4. What insights can be derived from age distribution?
5. How is a box plot different from a histogram?

4.10 Experiment No. 10 — Data Visualization III (Iris Dataset)

Aim:

To load the Iris dataset, analyse its features, and perform visualisations, including histograms and box plots, to identify distributions and outliers.

Objective:

To study:

1. Scanning datasets for feature types (numeric and categorical).
2. Plotting feature-wise histograms.
3. Creating box plots to identify outliers.
4. Comparing the distributions of features.

Theory:

The **Iris dataset** is a multivariate dataset containing 150 samples belonging to 3 flower species:

- Iris setosa
- Iris versicolor
- Iris virginica

Features include:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Histograms help visualize distribution (normal, skewed, uniform).
Box plots show the five-number summary and reveal outliers using IQR.

Applications:

- Outlier detection
- Feature behavior comparison
- Exploratory Data Analysis (EDA)

- Data-driven decision-making for ML model selection

Input:

Iris dataset downloaded from UCI ML Repository or loaded from sklearn.datasets.

Output:

- List of features with data types
- Histograms for all features
- Box plots with outlier visualization
- Inference comparing distributions

Conclusion:

Feature types were identified, and visualizations revealed distinct distribution patterns across features. Box plots highlighted outliers, especially in petal-related features, offering insights useful for model building.

Outcome (ELO):

- **ELO1:** Identify feature types in datasets.
- **ELO2:** Apply histogram and box plot visualizations.
- **ELO3:** Compare distributions and interpret outliers.

Viva Questions:

1. Why do we create histograms for each feature?
2. What is the difference between normal and skewed distribution?
3. What does a box plot represent?
4. What is IQR, and how are outliers detected?
5. Why is Iris dataset useful for ML experiments?

4.11 Experiment No. 11 — Working with Impala (Database Creation & Queries)

Aim:

To create databases and tables, insert data, and run basic queries using Apache Impala.

Objective:

To study:

1. Basics of Apache Impala for SQL processing.
2. Creating and describing databases and tables.
3. Inserting and retrieving data using SQL commands.
4. Understanding distributed query execution.

Theory:

Apache Impala is a massively parallel SQL engine for processing large-scale data stored in Hadoop Distributed File System (HDFS). It offers:

- Low-latency SQL queries
- Support for SELECT, INSERT, JOIN, AGGREGATIONS
- Integration with Hive Meta store
- Real-time analytics on big data

Common Impala commands include:

- CREATE DATABASE
- CREATE TABLE
- INSERT INTO
- SELECT
- DESCRIBE

Applications:

- Big Data analytics
- Real-time SQL query processing

- Large-scale reporting
- ETL pipelines using distributed SQL

Input:

Impala shell and sample data (e.g., student records or employee table).

Output:

- Successfully created database
- Created and populated tables
- Executed SELECT queries on inserted data

Conclusion:

A database and table were successfully created using Impala. Records were inserted and retrieved using SQL queries. Impala demonstrated fast and efficient distributed query execution.

Outcome (ELO):

- **ELO1:** Use Impala shell and commands.
- **ELO2:** Manage databases and tables in Impala.
- **ELO3:** Execute and interpret SQL queries in a Big Data environment.

Viva Questions:

1. What is Impala used for?
2. How is Impala different from Hive?
3. Why do we use HDFS with Impala?
4. What is the role of Hive Meta store in Impala?
5. What is parallel query execution?

4.12 Experiment No. 12 — Scala Program using Apache Spark

Aim:

To write and execute a simple Scala program using Apache Spark framework.

Objective:

To study:

1. Basics of Scala programming.
2. Running Scala code on Apache Spark.
3. Using RDDs or DataFrames for basic operations.

Theory:

Apache Spark is a distributed processing engine used for fast data analytics.
Scala is Spark's native language and enables:

- Functional + object-oriented programming
- Distributed data structures
- Lazy evaluation
- Fault-tolerant resilient distributed datasets (RDDs)

Basic Spark program flow:

1. Create Spark Session
2. Load or create data (RDD/DataFrame)
3. Apply transformations (map, filter, flatMap)
4. Perform actions (collect, count, show)

Example simple operations:

- Word count
- Filtering values
- Basic arithmetic transformations

Applications:

- Large-scale data processing
- Real-time analytics
- Batch processing
- Machine learning pipelines using Spark MLlib

Input:

Scala program executed via Spark Shell or sbt.

Output:

- Successful execution of Scala code
- Results printed on console
- Basic transformations and actions demonstrated

Conclusion:

A Scala program was written and executed on Apache Spark demonstrating distributed data transformations. This experiment provided understanding of Spark's architecture and functional programming in Scala.

Outcome (ELO):

- **ELO1:** Understand basics of Scala.
- **ELO2:** Execute Spark programs using Scala.
- **ELO3:** Apply transformations and actions on distributed datasets.

Viva Questions:

1. Why is Scala preferred for Spark?
2. What is an RDD?
3. What is the difference between transformation and action?
4. What is a Spark Session?
5. Give examples of Spark transformations.